Hylan Thomas

Present state y_2y_1	Next state w=0 Y_2Y_1	Next state w=1 Y_2Y_1	out z
00	10	11	0
01	01	00	0
10	11	00	0
11	10	01	1

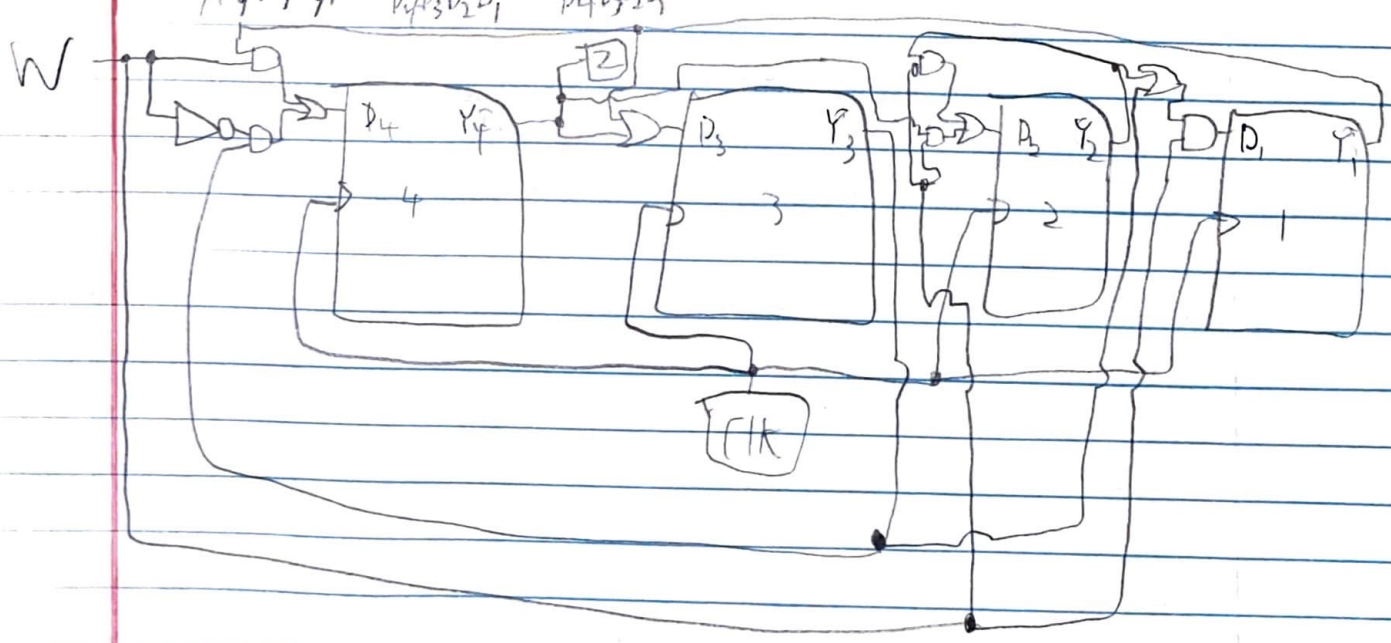
Present state	Next state w=0 Y_2Y_1	Next state w=1 Y_2Y_1	out z
A	C	D	0
B	B	A	0
C	D	A	0
D	C	B	1

P state $y_4y_3y_2y_1$	w=0 $D_4D_3D_2D_1$	w=1 $D_4D_3D_2D_1$	out z
0001	0100	1000	0
0010	0010	0001	0
0100	1000	0001	0
1000	0100	0010	1

$D_1 = w(y_2+y_3)$

$D_2 = \bar{w}y_2 + wy_4$

$D_3 = y_4 + y_1$

$D_4 = w(y_1) + \bar{w}y_3$

$y_4 = z$